

Question:

Write a JAVA program to analyze the performance of a student. Create a **Student** class with roll number, name and marks of 5 subjects, a constructor, and methods to calculate the total, average, highest, lowest, percentage, grade, result (pass/fail) and a performance remark. Read the details from the user in main(), apply string operations on the name, and display a complete performance report.

Execution Steps:

Step 1: Program execution starts from main().

Step 2: Scanner object sc is created.

Step 3: The heading "STUDENT PERFORMANCE ANALYSIS SYSTEM" is displayed.

Step 4: User enters the roll number 101 and sc.nextInt() stores it in rollNumber.

Step 5: sc.nextLine() consumes the leftover newline so the name can be read correctly.

Step 6: User enters the name "Shubhang Chourey" and sc.nextLine() stores it in studentName.

Step 7: An int array marks of size 5 is created and the loop reads the 5 subject marks: 88, 92, 79, 85, 90.

Step 8: Student object is created: new Student(rollNumber, studentName, marks). The constructor assigns the values using this.

Step 9: student.displayDetails() is called.

Step 10: The name is converted using trim() and toUpperCase(): "SHUBHANG CHOUREY", and length() gives 16.

Step 11: calculatePercentage() calls calculateTotal() ($88+92+79+85+90 = 434$) and computes $434 / 500 * 100 = 86.8$, rounded to two decimal places.

Step 12: determineGrade(86.8) goes through the if-else ladder: not ≥ 90 , but ≥ 80 , so the grade is A.

Step 13: The report is printed: total 434, average 86.80 ($434 / 5$), highest 92 (findHighest()) and lowest 79 (findLowest()).

Step 14: getResult(86.8) returns PASS because $86.8 \geq 50$, and getRemark("A") uses switch to return "Excellent Performance".

Step 15: Scanner is closed.

Step 16: Program ends.

Algorithm:

Step 1: Start.

Step 2: Create a Scanner object to read input.

Step 3: Display the heading of the system.

Step 4: Read the roll number and the student name.

Step 5: Create an array of size 5 and read the marks of 5 subjects using a for loop.

Step 6: Create a Student object using the parameterized constructor.

Step 7: Call displayDetails() on the object.

Step 8: Convert the name using trim() and toUpperCase() and find its length.

Step 9: Calculate the total (for-each loop) and the average (total / number of subjects).

Step 10: Find the highest and lowest marks using Math.max() and Math.min().

Step 11: Calculate the percentage = total / 500 * 100 and round it to two decimal places.

Step 12: Determine the grade: A+ (≥ 90), A (≥ 80), B (≥ 70), C (≥ 60), D (≥ 50), otherwise F.

Step 13: Determine the result: PASS if the percentage is ≥ 50 , otherwise FAIL.

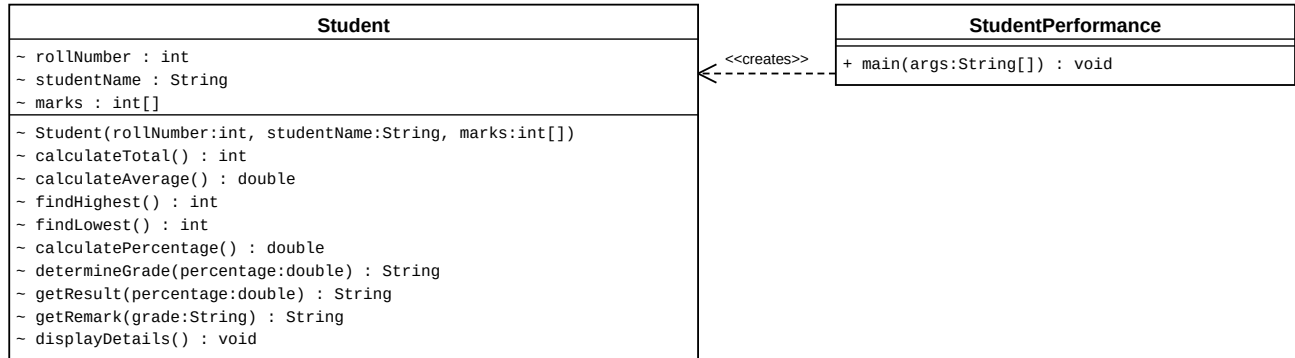
Step 14: Get the performance remark for the grade using switch.

Step 15: Display the complete performance report.

Step 16: Close the Scanner.

Step 17: Stop.

UML Class Diagram:



The dashed arrow shows that `main()` in `StudentPerformance` creates and uses a `Student` object. '~' means default (package) access, '+' means public.

Code:

```
import java.util.Scanner;

class Student {
    int rollNumber;
    String studentName;
    int[] marks;

    // Constructor to initialize student details
    Student(int rollNumber, String studentName, int[] marks) {
        this.rollNumber = rollNumber;
        this.studentName = studentName;
        this.marks = marks;
    }

    // Calculate total marks
    int calculateTotal() {
        int total = 0;

        for (int mark : marks) {
            total += mark;
        }

        return total;
    }

    // Calculate average marks
    double calculateAverage() {
        return (double) calculateTotal() / marks.length;
    }

    // Find highest mark
    int findHighest() {
        int highest = marks[0];

        for (int mark : marks) {
            highest = Math.max(highest, mark);
        }

        return highest;
    }

    // Find lowest mark
    int findLowest() {
        int lowest = marks[0];

        for (int mark : marks) {
            lowest = Math.min(lowest, mark);
        }

        return lowest;
    }

    // Calculate percentage
    double calculatePercentage() {
        double percentage = (double) calculateTotal() / 500 * 100;

        // Round percentage to two decimal places
        return Math.round(percentage * 100.0) / 100.0;
    }

    // Determine grade
    String determineGrade(double percentage) {
        if (percentage >= 90)
            return "A+";
        else if (percentage >= 80)
            return "A";
        else if (percentage >= 70)
            return "B";
    }
}
```

```

        else if (percentage >= 60)
            return "C";
        else if (percentage >= 50)
            return "D";
        else
            return "F";
    }

    // Determine pass or fail
    String getResult(double percentage) {
        if (percentage >= 50)
            return "PASS";
        else
            return "FAIL";
    }

    // Performance remark
    String getRemark(String grade) {
        switch (grade) {
            case "A+":
                return "Outstanding Performance";
            case "A":
                return "Excellent Performance";
            case "B":
                return "Very Good Performance";
            case "C":
                return "Good Performance";
            case "D":
                return "Satisfactory Performance";
            default:
                return "Needs Improvement";
        }
    }
}

// Display complete performance report
void displayDetails() {

    // String operations
    String formattedName = studentName.trim().toUpperCase();
    int nameLength = formattedName.length();

    double percentage = calculatePercentage();
    String grade = determineGrade(percentage);

    System.out.println("\n=====");
    System.out.println("          STUDENT PERFORMANCE REPORT");
    System.out.println("=====");

    System.out.println("Roll Number      : " + rollNumber);
    System.out.println("Student Name     : " + formattedName);
    System.out.println("Name Length      : " + nameLength);

    System.out.print("Subject Marks    : ");
    for (int mark : marks) {
        System.out.print(mark + " ");
    }

    System.out.println("\nTotal Marks      : " + calculateTotal());
    System.out.printf("Average Marks    : %.2f\n", calculateAverage());
    System.out.println("Highest Mark     : " + findHighest());
    System.out.println("Lowest Mark      : " + findLowest());
    System.out.println("Percentage       : " + percentage + "%");
    System.out.println("Grade            : " + grade);
    System.out.println("Result           : " + getResult(percentage));
    System.out.println("Performance Remark: " + getRemark(grade));

    System.out.println("=====");
}
}

```

```

public class StudentPerformance {
    public static void main(String[] args) {

        Scanner sc = new Scanner(System.in);

        System.out.println("=====");
        System.out.println("    STUDENT PERFORMANCE ANALYSIS SYSTEM");
        System.out.println("=====");

        System.out.print("Enter Roll Number: ");
        int rollNumber = sc.nextInt();

        sc.nextLine(); // Consume newline

        System.out.print("Enter Student Name: ");
        String studentName = sc.nextLine();

        int[] marks = new int[5];

        System.out.println("Enter marks for 5 subjects:");

        for (int i = 0; i < 5; i++) {
            System.out.print("Subject " + (i + 1) + ": ");
            marks[i] = sc.nextInt();
        }

        // Create Student object
        Student student = new Student(rollNumber, studentName, marks);

        // Display performance report
        student.displayDetails();

        sc.close();
    }
}

```

Output:

```
=====
      STUDENT PERFORMANCE ANALYSIS SYSTEM
=====
Enter Roll Number: 101
Enter Student Name: Shubhang Chourey
Enter marks for 5 subjects:
Subject 1: 88
Subject 2: 92
Subject 3: 79
Subject 4: 85
Subject 5: 90

=====
      STUDENT PERFORMANCE REPORT
=====
Roll Number      : 101
Student Name     : SHUBHANG CHOUREY
Name Length      : 16
Subject Marks    : 88 92 79 85 90
Total Marks      : 434
Average Marks    : 86.80
Highest Mark     : 92
Lowest Mark      : 79
Percentage       : 86.8%
Grade           : A
Result          : PASS
Performance Remark: Excellent Performance
=====
```